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HW 6 - Bayesian Inference

Bayes' Theorem

$$P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}$$

Binomial Prob Dist/Likelihood

$$P(h|\theta) = \binom{n}{h} \theta^h (1-\theta)^{(n-h)}$$

β - Fxn / Prior

$$P(\alpha, \beta, \theta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{(\alpha-1)}(1-\theta)^{(\beta-1)}$$

1. 7 flips, 2 heads, $\beta(2,5)$

A.) Now gets 3 heads + 2 tails

$$P(\theta|E) = \frac{\text{Likelihood w/ } \beta(3,2) \times \text{Beta-fxn w/ } \beta(2,5)}{P(E) = \int_0^1 P(\theta|H)\, P(H)\, d\theta}$$

$$= \frac{\binom{5}{3}\theta^3(1-\theta)^2 \cdot \frac{\Gamma(7)}{\Gamma(2)\Gamma(5)} \theta(1-\theta)^4}{\int_0^1 \underbrace{\binom{5}{3} \frac{\Gamma(7)}{\Gamma(2)\Gamma(5)}}_{\text{Const}} \theta^4(1-\theta)^6\, d\theta}$$

$$\int_0^1 \theta^4(1-\theta)^6\, d\theta = \frac{\Gamma(4+1)\Gamma(6+1)}{\Gamma(4+1+6+1)}$$ Beta-Fxn Property

$$\Rightarrow P(\theta|E) = \frac{\cancel{\binom{5}{3} \frac{\Gamma(7)}{\Gamma(2)\Gamma(5)}} \theta^4(1-\theta)^6}{\cancel{\binom{5}{3} \frac{\Gamma(7)}{\Gamma(2)\Gamma(5)}} \frac{\Gamma(5)\Gamma(7)}{\Gamma(12)}}$$

$$\Rightarrow \boxed{P(\theta|E) = \frac{\Gamma(5+7)}{\Gamma(5)\Gamma(7)} \theta^4(1-\theta)^6 = \beta(5,7)}$$

B.) Binomial Coeff. for ${}_4C_2$

$$\binom{4}{2} = \frac{4!}{2!(4-2)!} = \frac{24}{4} = 6$$

C.) Beta Binomial Predictive Dist

$$P(k|n,\alpha,\beta) = \binom{n}{k} \frac{\Gamma(k+\alpha)\Gamma(n-k+\beta)}{\Gamma(n+\alpha+\beta)} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}$$

$$P(4|2,5,7) = 6 \frac{\Gamma(7)\Gamma(9)}{\Gamma(16)} \frac{\Gamma(12)}{\Gamma(5)\Gamma(7)}$$

$$\Gamma(x) = (x-1)!$$

$$= 6 \frac{\cancel{6!}\,8!}{15!} \frac{11!}{4!\,\cancel{6!}} = 0.307 = 30.7\%$$